Caleb Standfield

Salt Lake City, UT

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Education

University of Utah Salt Lake City, UT

Bachelor of Science in Computer Science | Artificial Intelligence Track | Minor in Japanese | Data Science Certificate

Expected Dec 2026; GPA: 3.788

Relevant Coursework: Algorithms, Data Structures, Computer Systems, Computer Organization, Software Practice I & II, Artificial Intelligence, Machine Learning, NLP, Data Mining, Linear Algebra, Discrete Mathematics

Technical Skills

Languages: Rust, C++, C#, Python, Java, TypeScript, SQL, HTML, CSS
Frameworks: Axum, Tokio, FastAPI, React Native, Next.js, Flutter, TailwindCSS, PyTorch, WebAssembly, Blazor
Tools: PostgreSQL, MongoDB, Docker, Git, Linux Command Line, Tree-sitter, LaTeX
Engineering: Backend Development, Full-Stack Development, Concurrency, API Design, System Design, ML Integration

Experience

Adobe May 2026 – Aug 2026

Software Engineer Intern

- Built an end-to-end LLM code migration pipeline in Python and TypeScript that converts AppMeasurement analytics code to the Web SDK, extending support beyond Adobe Tags to standalone JavaScript implementations.
- Made the pipeline source-agnostic, so JavaScript files, code snippets, and Adobe Tags custom code share the same ingest, translation, and writeback stages, with output as snippets, files, or zip archives.
- Used Tree-sitter, dependency graphs, and topological sorting to find analytics code and its context, then combined deterministic processing with LLM translation to generate native Web SDK code.
- Built frontend and backend review workflows that surfaced conversion results and pointed users to the exact code sections needing manual remediation.

Projects

Cadenza – *Capstone Project* Aug 2025 – Present

Rust / TypeScript / React Native / PostgreSQL

- Co-developed an Apple Music companion app for song tagging, query-based playlists, and listening analytics, and contributed to system architecture and caching across React Native and Rust services.
- Owned the LLM song-tagging integration end to end, connecting backend generation services to mobile workflows for reviewing and applying suggested tags.
- Built a persistent listening-event queue and transactional backend ingestion with deduplication, allowing failed requests to retry without double-counting listening events.
- Developed SQL-backed listening analytics with interactive calendar heatmaps, tag filters, and ranked listening history; contributed cached views for offline access.
- Built mobile navigation and playback interfaces modeled on Apple Music interaction patterns and contributed to native MusicKit integration.

Multithreaded Web Server – *Rust* Sep 2025 – Present

- Built a multithreaded HTTP server from scratch in Rust, using concurrency primitives and thread pooling for requests.
- Implemented asynchronous I/O and low-latency static file serving with a focus on low-level performance optimization.
- Currently conducting load tests on physical hardware to evaluate server performance under concurrent load.

Blackjack Game Engine – *C++ / Qt* Mar 2025 – Apr 2025

- Developed a complete blackjack game engine from scratch in C++ and Qt, including AI-controlled opponents.
- Used object-oriented design and a finite state machine to manage game state transitions and event handling.
- Collaborated on a six-person team, following agile development practices with daily scrum meetings.

Portfolio Website – *Next.js / TypeScript / TailwindCSS* Oct 2024 – Nov 2025

- Designed and built a responsive, minimal personal portfolio site to showcase my projects and experience.
- Developed custom canvas visuals and animations within a responsive layout for desktop and mobile.
- Optimized for performance and accessibility, and handled deployment end to end at calebstandfield.dev.